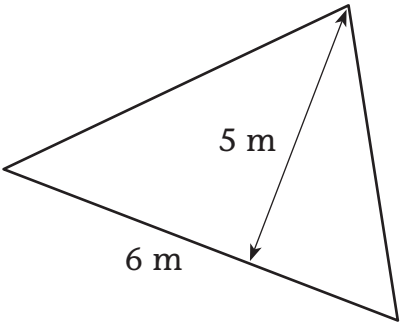
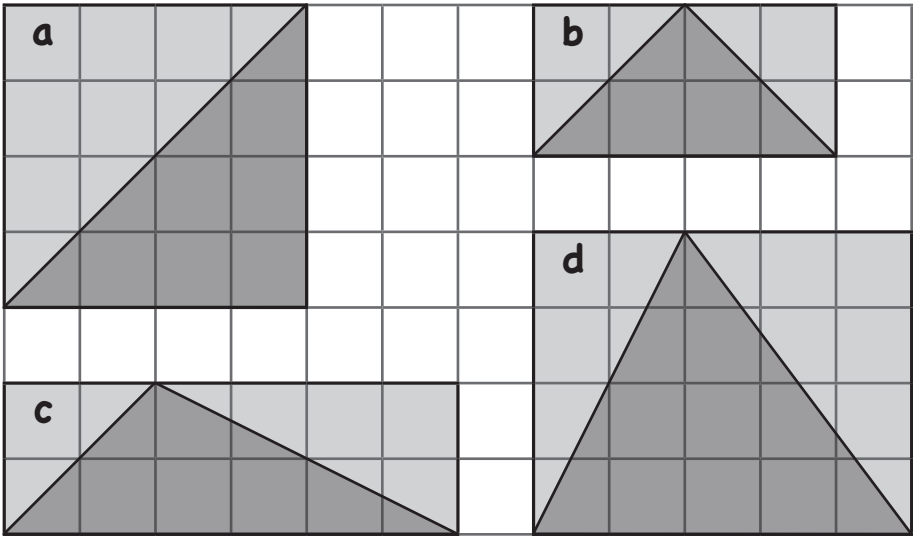


2.3 Areas of triangles

A piece of shade cloth is the shape of a triangle.
Darren wants to calculate its area.

- 1 In these diagrams, a triangle has been drawn inside each rectangle. The rectangles are drawn on a one centimetre grid. Find the area of each rectangle and the area of each triangle. Write your answers in the table provided.



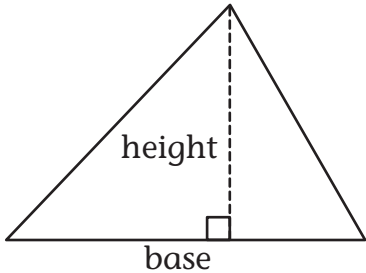
	Area of rectangle	Area of triangle
a		
b		
c		
d		

- 2 Compare the areas of each pair of triangles and rectangles in the table.
Complete this sentence:
The area of each triangle is _____ the area of the matching rectangle.

Therefore to calculate the area of a triangle,
we need the length of the base and the height.

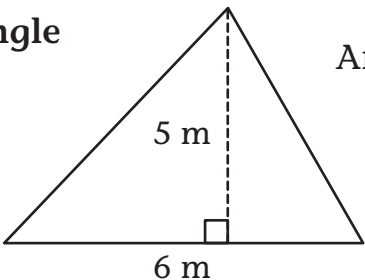
$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

Any side of a triangle can be chosen as the base.
The height must be at **right angles** to the chosen base.



EXAMPLE

The base of a triangle
is 6 m long.
The height is 5 m.
What is the area
of this triangle?

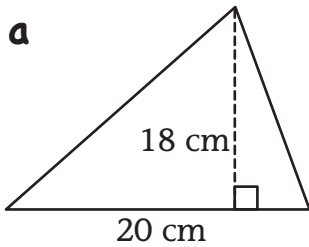


$$\begin{aligned} \text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2} \times 6 \times 5 \\ &= 15 \text{ m}^2 \end{aligned}$$

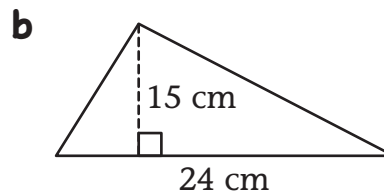
2.3 Areas of triangles

CONTINUED

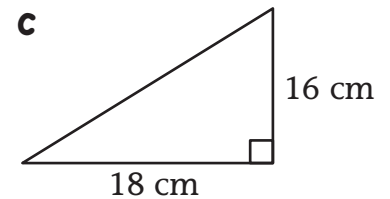
- 3** Use the rule $\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$ to calculate the area of each triangle.



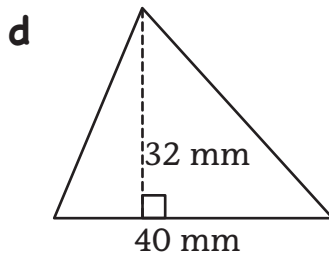
Area = _____
= _____



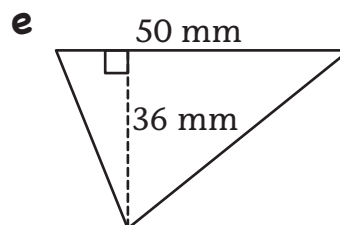
Area = _____
= _____



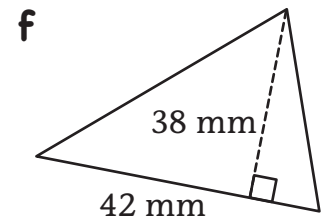
Area = _____
= _____



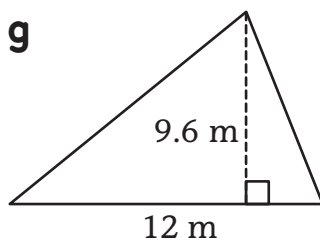
Area = _____
= _____



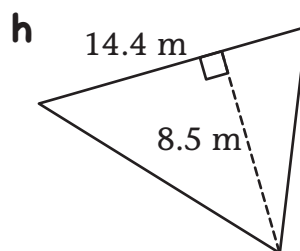
Area = _____
= _____



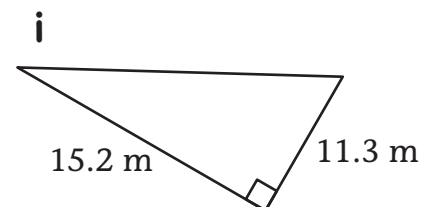
Area = _____
= _____



Area = _____
= _____



Area = _____
= _____

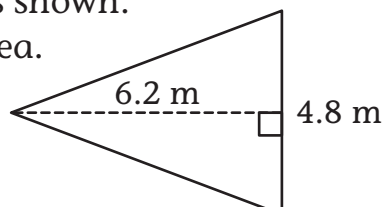


Area = _____
= _____

Answer the following questions in your workbook.

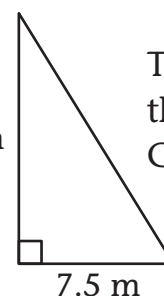
- 4** The base of a triangle measures 150 mm. Its height is 90 mm. Calculate the area of this triangle.

- 5** A piece of shade cloth is the shape of a triangle, as shown. Calculate its area.



6

18.7 m



The diagram shows the sail of a yacht. Calculate its area.